Hints: Number theory – 7

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Count the terms $\leqslant N$: there are $\lfloor N/\alpha \rfloor + \lfloor N/\beta \rfloor$ of them, roughly. Since this must equal N, dividing by N and letting $N \to \infty$ gives $\frac{1}{\alpha} + \frac{1}{\beta} = 1$; in particular $\alpha, \beta > 1$. If $\alpha = p/q$ were rational, then $\beta = p/(p-q)$, and $\lfloor q\alpha \rfloor = \lfloor (p-q)\beta \rfloor = p$ would collide.

Hint 2. Non-squares: m is the n-th non-square if and only if $n = m - \lfloor \sqrt{m} \rfloor$. Put $k = \lfloor \sqrt{m} \rfloor$; then $k^2 < m \leqslant k^2 + 2k$, so $k^2 - k < n \leqslant k^2 + k$. This is equivalent to $k - \frac{1}{2} < \sqrt{n} < k + \frac{1}{2}$, i.e. $k = \lfloor \sqrt{n} + \frac{1}{2} \rfloor$.

Hint 3. Since $\frac{1}{\sqrt{2}} + \frac{1}{2+\sqrt{2}} = 1$, Rayleigh–Beatty says that the positive integers not of the form $\lfloor k\sqrt{2} \rfloor$ are exactly the numbers $\lfloor k(2+\sqrt{2}) \rfloor$.

Problem 2

Hint 1. First identity: $(\sqrt{n} + \sqrt{n+1})^2 = 2n + 1 + 2\sqrt{n^2+n}$ lies strictly between $4n+1$ and $4n+2$. No perfect square is $\equiv 2, 3 \pmod 4$.

Hint 2. Second: let s be the sum of the three cube roots. By concavity, $s < 3\sqrt[3]{n+1}$. By AM–GM, $s > 3\big(n(n+1)(n+2)\big)^{1/9}$, and $n(n+1)(n+2) > (n + \frac{26}{27})^3$ for $n \geqslant 9$. Check $n \leqslant 8$ directly.

Hint 3. So $27n+26 < s^3 < 27n+27$. An integer $K \leqslant s$ has $K^3 < 27n+27$, hence $K^3 \leqslant 27n+26$. Therefore $\lfloor s \rfloor = \lfloor \sqrt[3]{27n+26} \rfloor$.

Problem 3

Hint 1. Put $N = 10^{100}$, so the number is $\lfloor N^{200}/(N+3) \rfloor$. Divide N^{200} by $N+3$ as polynomials in N.

Hint 2. $N^{200} - 3^{200} = (N+3)\sum_{j=0}^{199} N^{199-j}(-3)^j$, and $0 < 3^{200} < N+3$. So the integer part is $\sum_{j=0}^{199} N^{199-j}(-3)^j$.

Hint 3. Modulo 10 only the term $j = 199$ survives: $-3^{199} \equiv -7 \equiv 3$. Answer: 3.

Problem 4

Hint 1. Among $m, m+1, \ldots, n$ exactly one number has the largest power of 2, say 2^t with $t \geqslant 1$. (Between two multiples of 2^t there is a multiple of 2^{t+1}.) In the sum, that term alone has 2-adic valuation $-t$, so the sum has valuation $-t < 0$.

Hint 2. The denominator of H_n divides $L(n)$, the common denominator. By Hint 1 with $m = 1$, v_2 of the denominator is $t = \lfloor \log_2 n \rfloor$. A prime $p \in (\frac{n}{2}, n]$ divides only the term $k = p$, so the same argument with p in place of 2 applies.

Hint 3. Example: $H_6 = \frac{49}{20}$ while $L(6) = 60$.

Problem 5

Hint 1. Since $\sum k a_k = 0$, you may subtract kt from $\lfloor kx \rfloor$ for any constant t without changing the sum. Choose t so that $\sum_{k \leqslant n}(\lfloor kx \rfloor - kt) = 0$, i.e. $t = C_n$ where $C_k = \frac{2B_k}{k(k+1)}$ and $B_k = \sum_{j \leqslant k} \lfloor jx \rfloor$.

Hint 2. Abel summation: $\sum_k a_k(\lfloor kx\rfloor - kC_n) = -\sum_{k<n}(a_{k+1}-a_k)\frac{k(k+1)}{2}(C_k - C_n)$. Since $a_{k+1} \geqslant a_k$, it suffices to prove that C_k is non-decreasing.

Hint 3. $C_k \leqslant C_{k+1}$ is equivalent to $2B_k \leqslant k\lfloor (k+1)x\rfloor$. Sum the superadditivity $\lfloor jx\rfloor + \lfloor (k+1-j)x\rfloor \leqslant \lfloor (k+1)x\rfloor$ over $j = 1, \ldots, k$.

Problem 6

Hint 1. Compare growth. The right side is $n - \frac{n}{x} + \frac{n}{x^2} - \cdots + O(\log n) = \frac{nx}{x+1} + O(\log n)$, and the left side is $\frac{n}{x} + O(1)$. So $x^2 = x + 1$, i.e. $x = \varphi = \frac{1+\sqrt{5}}{2}$.

Hint 2. Sufficiency: the right side $R(n)$ satisfies $R(n) = n - R(\lfloor n/x\rfloor)$, $R(0) = 0$. Prove by strong induction that $R(n) = \lfloor (n+1)/\varphi\rfloor$. This amounts to $\lfloor \frac{n+1}{\varphi}\rfloor + \lfloor \frac{\lfloor n/\varphi\rfloor + 1}{\varphi}\rfloor = n$.

Hint 3. To prove the last identity, use that $\lfloor k\varphi\rfloor$ and $\lfloor k\varphi^2\rfloor$ partition the positive integers (Rayleigh–Beatty with $\frac{1}{\varphi} + \frac{1}{\varphi^2} = 1$). Count how many terms of each sequence are $\leqslant n$ or $\leqslant n+1$.

Problem 7

Hint 1. Reduction: the order of σ is the lcm of its cycle lengths. Write this lcm as a product of prime powers $q_1 \cdots q_r$; each q_i divides some cycle length, so $\sum q_i \leqslant n$. Replacing the cycles by cycles of lengths $q_1, \ldots, q_r$ (plus fixed points) keeps the order.

Hint 2. Lower bound: use all primes $p \leqslant y = c\sqrt{n\ln n}$. Their sum is at most $y\pi(y) \leqslant Cy^2/\ln y \leqslant n$ for small c. Then $\ln g(n) \geqslant \theta(y) \geqslant c'y$ by Chebyshev.

Hint 3. Upper bound with $y = \sqrt{n\ln n}$: factors $q_i \leqslant y$ involve distinct primes $\leqslant y$, so there are at most $\pi(y)$ of them, each contributing $\ln q_i \leqslant \ln n$. Factors $q_i > y$ number at most n/y, each contributing $\leqslant \ln n$. Both totals are $O(\sqrt{n\ln n})$.

Problem 8

Hint 1. Let $\alpha = \sqrt{2} - 1$, $\beta = 1/\alpha = 2 + \alpha$, $S(n) = \sum_{k\leqslant n}(-1)^{\lfloor k\alpha\rfloor}$. Group k by $j = \lfloor k\alpha\rfloor$: block j consists of $\lfloor j\beta\rfloor < k \leqslant \lfloor (j+1)\beta\rfloor$ and has length $2 + \lfloor (j+1)\alpha\rfloor - \lfloor j\alpha\rfloor$.

Hint 2. Summing over complete blocks: $S(\lfloor J\beta\rfloor) = 2[J \text{ odd}] + \sum_{j<J}(-1)^j(\lfloor (j+1)\alpha\rfloor - \lfloor j\alpha\rfloor)$. The last sum has a term exactly when $j + 1 = \lceil m\beta\rceil$, with sign $(-1)^{\lceil m\beta\rceil - 1} = (-1)^{\lfloor m\alpha\rfloor}$. Hence $S(\lfloor J\beta\rfloor) = 2[J \text{ odd}] + S(\lfloor J\alpha\rfloor)$.

Hint 3. Strong induction on n. If n lies in block J with J even, the partial block only adds, so $S(n) \geqslant S(\lfloor J\beta\rfloor) \geqslant 0$. If J is odd, the partial block subtracts, so $S(n) \geqslant S(\lfloor (J+1)\beta\rfloor) = S(\lfloor (J+1)\alpha\rfloor) \geqslant 0$. Check that $\lfloor (J+1)\alpha\rfloor < n$.

Problem 9

Hint 1. If $\alpha \leqslant 1$, then $S(\alpha)$ already contains every positive integer. So $\alpha, \beta, \gamma > 1$, and put $\theta_i = 1/\alpha, 1/\beta, 1/\gamma$. Exactly $\lceil n\theta\rceil - 1$ elements of $S(1/\theta)$ lie in $[1, n-1]$.

Hint 2. The partition gives $\sum_i(\lceil n\theta_i\rceil - 1) = n - 1$ for all n. Dividing by n gives $\sum\theta_i = 1$, and then $\sum_i\{-n\theta_i\} = 2$ for every $n \geqslant 1$.

Hint 3. By simultaneous Dirichlet approximation (pigeonhole), find $q \geqslant 2$ with $\|q\theta_i\| < \varepsilon$ for all i; this works for rational θ_i too. For $n = q - 1$, $\{-n\theta_i\} = \{\theta_i - q\theta_i\}$ is within ε of θ_i. So $\sum_i\{-n\theta_i\}$ is close to $\sum\theta_i = 1$, not 2.

Problem 10

Hint 1. Write $A_n = \sum_{k=0}^n \frac{n!}{k!}$, so the sum is $A_n/n!$ and $A_n = nA_{n-1} + 1$. Then $g(n) = n!/d_n$ with $d_n = \gcd(n!, A_n)$. The answer is *yes*: show that d_n cannot be large for many consecutive n.

Hint 2. For $m < n$, $\gcd(A_m, A_n)$ divides $A_n - \frac{n!}{m!}A_m = \sum_{k=m+1}^{n} \frac{n!}{k!} \leqslant e\, n^{n-m-1}$. So the d_j for nearby j are almost coprime.

Hint 3. Suppose $g(j) \leqslant j^{0.999j}$, i.e. $d_j \geqslant j!/j^{0.999j}$, for all $j \in [n, n+K)$. Use $\prod_j d_j \leqslant \operatorname{lcm}(d_j) \prod_{i<j} \gcd(d_i, d_j) \leqslant (n+K)!\,(e(n+K)^K)^{K^2}$.

Hint 4. The assumed lower bound gives $\prod_j d_j \geqslant \left(n^{0.001n}e^{-n}\right)^K$ roughly. With $K = 2000$ and n huge, this exceeds the upper bound (compare $2n\ln n$ with $n\ln n$). So some j has $g(j) > j^{0.999j}$.